

Control work 6 — the circle, and work on the mistakes

Circumference, area, sectors and a compound shape, under time.

LESSON 109–110

2 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 6

S7 8 AND 15 REVIEW

LESSON CLOCK

3

35'

12'

25'

5'

Instructions	3 min
The paper	35 min
Answers	12 min
Diagnosis and rewrite	25 min
The map	5 min

LEARNING OBJECTIVES

- Use the two circle formulae accurately and with units.
- Work backwards from a circumference or an area.
- Find a sector area and a semicircle perimeter.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 250–318
Cambridge	Stage 7, pp. 80–84, 148–156
Workbook	Control paper 6

01

Explanation

The paper — 25 marks, 35 minutes

Q	TASK	MARKS	FROM
1	Name the centre, a radius, a diameter and a chord on a given circle	3	L89–91
2	Find C and S for a circle of radius 14 cm	4	L92–99
3	A circle has $C = 66$ cm: find its radius	3	L92–94
4	A disc has area 154 cm ² : find its radius	3	L97–99
5	Find the area and perimeter of a semicircle of radius 7 cm	4	L95–102
6	Find the area of a 120° sector of radius 21 cm	4	L100–102
7	A semicircle of radius 7 sits on a 14 × 10 rectangle: find the area	4	L103–105

THE ANSWERS

88 cm and 616 cm²; 10.5 cm; 7 cm; 77 cm² and 36 cm; 462 cm²; 217 cm².

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
diameter used for radius	$\pi \cdot 28^2$	$\pi \cdot 14^2$
formulae swapped	$C = \pi r^2$	$C = 2\pi r$
square root forgotten	$r = 49$	$r = 7$
semicircle perimeter without the diameter	22	36
sector fraction wrong	$\frac{120}{180}$	$\frac{120}{360}$
compound shape: pieces not added	140	$140 + 77$
units missing	616	616 cm^2

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

IDEA	FORMULA OR RULE
circumference	$C = 2\pi r = \pi d$
area of a disc	$S = \pi r^2$
arc	$\frac{\alpha}{360} \cdot C$
sector area	$\frac{\alpha}{360} \cdot \pi r^2$
semicircle perimeter	$\pi r + 2r$
quadrant perimeter	$\frac{1}{2}\pi r + 2r$
compound shapes	add or subtract; trace the boundary

LOOKING FORWARD

Next come two Cambridge lessons on constructions and symmetry, and then the largest block of the quarter: speed, distance and time.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $C = 66$ cm.

$$① \quad C = 2\pi r$$

$$② \quad 66 = 2 \cdot \frac{22}{7} \cdot r$$

$$③ \quad 66 = \frac{44}{7} r$$

$$④ \quad r = 10.5 \text{ cm.}$$

$$\text{Check: } 2 \cdot \frac{22}{7} \cdot 10.5 = 66 \checkmark$$

ANSWER

10.5 cm

EXAMPLE 2 Model answer, Q5: a semicircle of radius 7 cm.

① Area: $\frac{1}{2} \cdot 154 = 77 \text{ cm}^2$.

② Curve: $\pi r = 22 \text{ cm}$.

③ Perimeter: $22 + 14 = 36 \text{ cm}$.

The diameter is part of the boundary.

ANSWER

77 cm^2 and 36 cm

EXAMPLE 3 Model answer, Q7: the compound shape.

① Rectangle: 140 cm^2 .

② Semicircle: 77 cm^2 .

③ $140 + 77 = 217 \text{ cm}^2$.

ANSWER

217 cm^2

03 Interactive model

Mark Q5 in two halves — area and perimeter — and report them separately; the class sees which of the two it actually finds hard.

The chapter in eight questions

C for $r = 14$:

44

88

616

196

S for $r = 14$:

88

196

308

616

$C = 66$ gives r :

10.5

21

7

33

$S = 154$ gives r :

7

49

14

11

A semicircle of radius 7 has area:

154

77

44

36

And perimeter:

22

36

44

77

A 120° sector of radius 21:

462

1386

44

132

The compound shape of Q7 has area:

140

77

217

294

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Circumference	Aylana uzunligi	Длина окружности
Area	Yuza	Площадь
Sector	Sektor	Сектор
Semicircle	Yarim doira	Полукруг
Compound shape	Murakkab shakl	Составная фигура
Units	O'lchov birligi	Единицы
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q2 needs two formulae because it asks for:

two circles

a length and an area

two radii

a check

Q3 divides by:

π

2π

r

2

Q4 ends with:

a division

a square root

a multiplication

a subtraction

Q5 perimeter includes:

the curve only

the curve and the diameter

two radii

the whole circle

Q6 fraction is:

$$\frac{120}{180}$$

$$\frac{120}{360}$$

$$\frac{360}{120}$$

$$\frac{1}{2}$$

Q7 is answered by:

subtracting

adding two pieces

tracing

measuring

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 14$: C

ANSWER 88 cm

2. $r = 14$: S

ANSWER 616 cm^2

3. $C = 66$: the radius

ANSWER 10.5 cm

4. $S = 154$: the radius

ANSWER 7 cm

5. A semicircle of radius 7: the area

ANSWER 77 cm^2

6. And the perimeter

ANSWER 36 cm

7. A 120° sector of radius 21

ANSWER 462 cm^2

Medium · the standard question

7 PROBLEMS

1. A semicircle of radius 7 on a 14×10 rectangle

ANSWER 217 cm^2

2. Its perimeter

ANSWER 56 cm

3. $r = 3.5$: C and S

ANSWER $22 \text{ cm}, 38.5 \text{ cm}^2$

4. A quadrant of radius 14: the area

ANSWER 154 cm^2

5. And its perimeter

ANSWER 50 cm

6. $C = 44$: the area of the disc

ANSWER 154 cm^2

7. $S = 616$: the circumference

ANSWER 88 cm

Hard · stretch and reason

7 PROBLEMS

1. A ring between radii 14 and 21

ANSWER 770 cm^2

2. A 45° sector of radius 28

ANSWER 308 cm^2

3. A square of side 28 less a disc of radius 14

ANSWER 168 cm^2

4. A 270° sector of radius 14: area and perimeter

ANSWER 462 cm^2 , 94 cm

5. Doubling the radius: C and S

ANSWER Doubled and quadrupled

6. A track 80×70 with ends of radius 35: one lap

ANSWER 380 m

7. Why does an area answer need cm^2 ?

ANSWER It measures two directions

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on, with its units.

1. Rewrite in full every question on which you lost a mark.
2. Find C and S for a circle of radius 10.5 cm .
3. A disc has area 616 cm^2 . Find its radius and circumference.
4. Find the area and perimeter of a quadrant of radius 21 cm .
5. Find the area of a 150° sector of a circle of radius 14 cm .